

Chapter 6: Symmetries

Quantum Mechanics

1 Introduction to Symmetry

Definition 1.1 (Symmetry). [def:symmetry] A *symmetry* of a system is some transformation, such that the system is invariant before and after the transformation.

Definition 1.2 (Symmetry of Quantum System). [def:symmetry-qm] A *symmetry* of a quantum system is a transformation that leaves the Hamiltonian invariant.

Definition 1.3 (Types of Symmetry). [def:symmetry-types] There are several types of symmetries in quantum mechanics, including:

- *Translation symmetry*: The system is invariant under some spatial translations.
- *Rotation symmetry*: The system is invariant under some rotations.
- *Reflection symmetry*: The system is invariant under reflections on some axis/planes.
- *Inversion symmetry*: The system is invariant under inversion of all coordinates.
- *Time-reversal symmetry*: The system is invariant under the reversal of time.

Theorem 1.1 (Symmetry Condition on the Hamiltonian). [thm:symmetry-condition-hamiltonian] $\hat{U}$ is a definition 1.2 of $\hat{H}$ iff

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H}.$$

Corollary 1.1 (Commutator Form). [cor:commutator-form-symmetry] For unitary $\hat{U}$,

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H} \iff \text{comm}(\hat{U}, \hat{H}) = 0.$$

Proof. Left-multiply by $\hat{U}$ and use $\hat{U}\hat{U}^\dagger = \hat{I}$:

$$\hat{U}^\dagger \hat{H} \hat{U} = \hat{H} \iff \hat{H} \hat{U} = \hat{U} \hat{H} \iff \text{comm}(\hat{U}, \hat{H}) = 0.$$

□

Remark 1.1 (“ $\hat{A}$ Obtained After Applying $\hat{U}$ ”). [rem:conjugation-vocabulary] The phrase “ $\hat{A}$ obtained after applying $\hat{U}$ ” refers to the conjugation $\hat{U}^\dagger \hat{A} \hat{U}$, since this preserves the expectation value.

2 Groups

Definition 2.1 (Group). [def:group] A *group* consists of the following data and assumptions:

- A set G .
- A binary operation $\cdot : G \times G \rightarrow G$ (the codomain being G bakes in closure).
- An identity element $e \in G$.
- An inverse function $(\cdot)^{-1} : G \rightarrow G$, written $g \mapsto g^{-1}$.
- *Associativity*: $(g_1 \cdot g_2) \cdot g_3 = g_1 \cdot (g_2 \cdot g_3)$ for all $g_1, g_2, g_3 \in G$.
- *Identity axiom*: $e \cdot g = g \cdot e = g$ for all $g \in G$.
- *Inverse axiom*: $g \cdot g^{-1} = g^{-1} \cdot g = e$ for all $g \in G$.

A collection with a well-behaved multiplication and inverses.

Example 2.1 (Real Numbers Under Addition). [ex:reals-group] Choose:

- Set $G := \mathbb{R}$ (the real numbers).
- Operation $\cdot := +$ (ordinary addition).
- Identity $e := 0$.
- Inverse $a^{-1} := -a$.

Associativity, identity, and inverse axioms follow from the usual properties of addition.

Example 2.2 (Boolean Set Under XOR). [ex:boolean-group] Choose:

- Set $G := \{\text{tt}, \text{ff}\}$ (equivalently $\{0, 1\}$).
- Operation $\cdot := \oplus$ (XOR, equivalent to addition mod 2).
- Identity $e := \text{ff}$ (equivalently 0).
- Inverse $g^{-1} := g$ (each element is self-inverse).

All axioms hold by direct verification.

Definition 2.2 (Continuous Group). [def:continuous-group] A group is *continuous* (a *Lie group*) if its elements are parametrized by a [continuous parameter space](#), and multiplication and inversion vary smoothly with the parameters. The real numbers under addition is an example.

Definition 2.3 (Discrete Group). [def:discrete-group] A group is *discrete* if its elements form a [discrete set](#) (finite or [countable](#)), with no continuous parameter connecting distinct elements. The Boolean set under XOR is an example.

Definition 2.4 (Continuous Symmetry). [def:continuous-symmetry] A symmetry is *continuous* if its operators form a continuous group.

Definition 2.5 (Discrete Symmetry). [def:discrete-symmetry] A symmetry is *discrete* if its operators form a discrete group.

3 Translation Operator

Definition 3.1 (Translation Operator). [def:translation-operator] Intuitively, a *translation operator* $\hat{T}(\vec{r}_0)$ (translates a state by a vector $\vec{r}_0$) acts on a state $|\psi\rangle$ should let $|\psi\rangle$ be translated by a vector $\vec{r}$, i.e.

$$\hat{T}(\vec{r}_0) |\psi\rangle = |\psi(\vec{r} - \vec{r}_0)\rangle$$

Definition 3.2 (Translation Operator (1D)). [def:translation-operator-1d] In one dimension, the translation operator $\hat{T}(a)$ shifts a state by a real distance $a \in \mathbb{R}$. Its action on a position-space wavefunction is

$$\hat{T}(a) \psi(x) = \psi(x - a).$$

Theorem 3.1 (Translation Operators Form a Symmetry Group). [thm:translation-symmetry-group] Suppose $\hat{T}(a)$ is a symmetry operator of a Hamiltonian $\hat{H}$ for every $a \in \mathbb{R}$. Then the collection of translation operators forms a group under operator composition:

- Set $G := \{\hat{T}(a) : a \in \mathbb{R}\}$.
- Operation $\cdot$: operator composition, $\hat{T}(a) \cdot \hat{T}(b) = \hat{T}(a + b)$.
- Identity $e := \hat{T}(0) = \hat{I}$.
- Inverse $\hat{T}(a)^{-1} := \hat{T}(-a)$.

All axioms follow from addition on $\mathbb{R}$.

Proof.

□

Theorem 3.2 (Exponential Form of the Translation Operator). [thm:translation-exponential] The one-dimensional translation operator can be written as:

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar}$$

Proof. Expand the exponential in a Taylor series at 0

$$\hat{T}(a) = e^{-ia\hat{p}/\hbar} = \sum_{n=0}^{\infty} \frac{(-ia/\hbar)^n}{n!} \cdot \hat{p}^n = \sum_{n=0}^{\infty} \frac{(-ia\hat{p})^n}{n!\hbar^n}$$

According to the definition of the momentum operator, $\hat{p} = -i\hbar \frac{d}{dx}$, so we can substitute this into the series expansion:

$$\hat{T}(a) = \sum_{n=0}^{\infty} \frac{(-ia(-i\hbar \frac{d}{dx}))^n}{n!\hbar^n} = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!}$$

So,

$$\hat{T}(a)\psi(x) = \sum_{n=0}^{\infty} \frac{(-a(\frac{d}{dx}))^n}{n!} \psi(x) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x)$$

Then, we do a Taylor expansion of $\psi(x)$ centered at $x + a$:

$$\psi(x) = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (x - (x+a))^n = \sum_{n=0}^{\infty} \frac{\psi^{(n)}(x+a)}{n!} (-a)^n = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x+a)$$

Substituting $x \rightarrow x - a$ in the previous identity,

$$\psi(x-a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}((x-a)+a) = \sum_{n=0}^{\infty} \frac{(-a)^n}{n!} \psi^{(n)}(x).$$

Which means,

$$\hat{T}(a)\psi(x) = \psi(x-a)$$

□

Theorem 3.3 ($\hat{T}(a)$ is a Unitary Operator). [thm:translation-unitary] *The translation operator $\hat{T}(a)$ is a unitary operator, i.e. $\hat{T}^\dagger(a) = \hat{T}^{-1}(a) = \hat{T}(-a)$.*

Remark 3.1 (Intuition about). [rem:intuition-about]

Proof. **TODO:** Proof to be written after clarifying preferred approach with the lecturer — direct from the action definition (definition 3.2), or via the exponential form (theorem 3.2). Email sent September 23, 2026. □

Theorem 3.4 (Position Operator After Applying Translation). [thm:position-under-translation] *Applying $\hat{T}(a)$ to the position operator $\hat{x}$ (see remark 1.1) gives*

$$\hat{T}(a)^\dagger \hat{x} \hat{T}(a) = \hat{x} + a\hat{I}.$$

Proof. Define $\hat{x}' := \hat{T}(a)^\dagger \hat{x} \hat{T}(a)$, and consider its action on a wavefunction $\psi(x)$:

$$\hat{x}'\psi(x) = \hat{T}(a)^\dagger \hat{x} \hat{T}(a) \psi(x)$$

Since operators are associative, we have

$$\hat{x}'\psi(x) = \hat{T}(a)^\dagger \hat{x} (\hat{T}(a) \psi(x)) = \hat{T}(a)^\dagger (\hat{x} \psi(x-a)) = \hat{T}(a)^\dagger (x \psi(x-a))$$

Treat $g(x) = \hat{x} \psi(x-a)$, so, $\hat{T}(a) g(x) = g(x) = x \hat{T}(a) \psi((x-a)-a)$.

Notice that $\hat{T}(a)$ is a unitary operator, ie $\hat{T}(a)^\dagger = \hat{T}(a)^{-1} = \hat{T}(-a)$, we have

$$\hat{x}'\psi(x) = \hat{T}(-a) (x \psi(x-a)) = (x+a) \psi((x+a)-a) = (x+a) \psi(x)$$

Rewrite the right-hand side in operator form:

$$(x+a) \psi(x) = x \psi(x) + a \psi(x) = \hat{x} \psi(x) + a\hat{I} \psi(x) = (\hat{x} + a\hat{I}) \psi(x).$$

So $\hat{x}' \psi = (\hat{x} + a\hat{I}) \psi$ for every ψ . Since two operators agreeing on all inputs are equal, we conclude

$$\hat{x}' = \hat{x} + a\hat{I}.$$

□

Theorem 3.5 (Momentum Operator After Applying Translation). [thm:momentum-under-translation] Applying $\hat{T}(a)$ to the momentum operator $\hat{p}$ (see remark 1.1) gives

$$\hat{T}(a)^\dagger \hat{p} \hat{T}(a) = \hat{p}.$$

Proof. Define $\hat{p}' := \hat{T}(a)^\dagger \hat{p} \hat{T}(a)$, and consider its action on a wavefunction $\psi(x)$:

$$\hat{p}'\psi(x) = \hat{T}(a)^\dagger \hat{p} \hat{T}(a) \psi(x) = \hat{T}(a)^\dagger \hat{p} \psi(x - a)$$

According to the definition of definition momentum operator position, $\hat{p} = -i\hbar d/dx$, plug in:

$$\hat{p}'\psi(x) = -\hat{T}(a)^\dagger i\hbar \frac{d\psi(x - a)}{dx}$$

Use $\hat{T}(a)^\dagger = \hat{T}(-a)$,

$$\hat{p}'\psi(x) = -\hat{T}(-a) i\hbar \frac{d\psi(x - a)}{dx} = -i\hbar \frac{d\psi(x)}{dx} = \hat{p}\psi(x)$$

So $\hat{p}'\psi = \hat{p}\psi$ for every ψ . Since two operators agreeing on all inputs are equal, we conclude

$$\hat{p}' = \hat{p}.$$

□

Theorem 3.6. For all operator, which only consists of $\hat{x}$ and $\hat{p}$, the conjugation action of $\hat{T}(a)$ on the operator is equivalent to replacing every $\hat{x}$ with $\hat{x} + a\hat{I}$ and leaving every $\hat{p}$ unchanged. ie,

$$Q(\hat{x}, \hat{p}) \longmapsto \hat{T}(a)^\dagger Q(\hat{x}, \hat{p}) \hat{T}(a) = Q(\hat{x} + a\hat{I}, \hat{p})$$

Theorem 3.7 (Translation Symmetry Condition on the Potential). [thm:translation-condition-potential] Let $\hat{H}$ be a 1D Hamiltonian. If $\hat{T}(a)$ is a symmetry operator of $\hat{H}$, then the potential is invariant under translation by a :

$$V(x + a) = V(x) \quad \text{for all } x.$$

Proof. Since $\hat{T}(a)$ is a unitary symmetry, corollary 1.1 gives

$$\hat{T}(a)^\dagger \hat{H} \hat{T}(a) = \hat{H} \iff \text{comm}(\hat{H}, \hat{T}(a)) = 0.$$

Notice that $\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x})$, and by theorem 3.5 and theorem 3.4, we have

$$\hat{T}(a)^\dagger \hat{H} \hat{T}(a) = \hat{T}(a)^\dagger \left(\frac{\hat{p}^2}{2m} + V(\hat{x}) \right) \hat{T}(a) = \frac{\hat{p}^2}{2m} + V(\hat{x} + a\hat{I}).$$

Which means, $\hat{T}(a)$ is a symmetry of $\hat{H}$ iff

$$V(x) = V(x + a)$$

□

Theorem 3.8 (Eigenvalues of the Translation Operator). [thm:translation-eigenvalues] The eigenvalues of the translation operator $\hat{T}(a)$ have modulus 1:

$$\hat{T}(a) |\psi\rangle = \lambda |\psi\rangle \implies \lambda = e^{i\theta} \quad \text{for some } \theta \in [0, 2\pi).$$

Proof. Since $\hat{T}(a)$ is unitary (theorem 3.3),

$$\langle \hat{T}(a) \psi \mid \hat{T}(a) \psi \rangle = \langle \psi \mid \hat{T}(a)^\dagger \hat{T}(a) \mid \psi \rangle = \langle \psi \mid \psi \rangle.$$

On the other hand,

$$\langle \hat{T}(a) \psi \mid \hat{T}(a) \psi \rangle = \langle \lambda \psi \mid \lambda \psi \rangle = |\lambda|^2 \langle \psi \mid \psi \rangle.$$

Equating and dividing by $\langle \psi \mid \psi \rangle \neq 0$:

$$|\lambda|^2 = 1 \implies \lambda = e^{i\theta} \quad \text{for some } \theta \in [0, 2\pi).$$

□

Theorem 3.9 (Bloch's Theorem). [thm:bloch-theorem] *If $T(a)$ is a symmetry of $\hat{H}$ for some $a \in \mathbb{R}$, stationary states of $\hat{H}$ have the form*

$$\psi(x) = e^{iqx} u(x)$$

Where

- q is called the crystal momentum
- $u(x)$ is periodic and have a period of a

Remark 3.2 (Stationary States may not be Periodic). [rem:stationary-states-may-not-be-periodic] From theorem 3.9, we can see that the stationary states $\psi(x)$ is not periodic. However $u(x)$ is periodic with period a .

Proof. Since $\hat{T}(a)$ is a symmetry of $\hat{H}$, we have $\text{comm}(\hat{H}, \hat{T}(a)) = 0$. According to theorem commuting share eigenvectors, stationary states of $\hat{H}$ can be chosen as simultaneous eigenvectors of $\hat{T}(a)$: $\hat{T}(a) \mid \psi \rangle = \lambda \mid \psi \rangle$.

By theorem 3.8, $\lambda = e^{i\theta}$ for some $\theta \in [0, 2\pi)$. Since $\hat{T}(a)$ may be written as $e^{-iap/\hbar}$, we identify $\theta = qa$ where $q = -p/\hbar$.

Then, we have

$$\hat{T}(a) \mid \psi \rangle = e^{iqa} \mid \psi \rangle \implies \psi(x - a) = e^{iqa} \psi(x)$$

Change $a \rightarrow -a$ to rewrite in the standard form:

$$\psi(x + a) = e^{iqa} \psi(x).$$

Define $u(x) := e^{-iqx} \psi(x)$. Then

$$u(x + a) = e^{-iq(x+a)} \psi(x + a) = e^{-iqx} \underbrace{e^{-iqa} e^{iqa}}_{=1} \psi(x) = u(x),$$

so u has period a , and rearranging gives $\psi(x) = e^{iqx} u(x)$.

Although $\hat{H}$ is translation-invariant, the state ψ is *not*:

$$\psi(x + a) = e^{iqa} \psi(x) \neq \psi(x).$$

However, the observable $|\psi|^2$ still respects the symmetry, since $|e^{iqa}| = 1$:

$$|\psi(x + a)|^2 = |e^{iqa}|^2 |\psi(x)|^2 = |\psi(x)|^2.$$

□

Theorem 3.10 (Continuous Translation Symmetry Forces Constant V). [thm:continuous-translation-constant-v] *If $\hat{T}(a)$ is a symmetry of $\hat{H}$ for every $a \in \mathbb{R}$, then V is constant.*

TODO

4 Vectors and Pseudovectors

5 Rotational Symmetry

6 Parity/Inverse Symmetry

Definition 6.1 (Parity Operator). [def:parity-operator] The *parity operator* Π (also called the *inversion operator*) is defined by

$$\Pi |\psi(\vec{r})\rangle = |\psi(-\vec{r})\rangle$$

Theorem 6.1 (Parity Squared is Identity). [thm:parity-squared] $\Pi^2 = \hat{I}$.

Proof. Applying Π twice to any ψ :

$$\Pi^2 \psi(\vec{r}) = \Pi \psi(-\vec{r}) = \psi(-(-\vec{r})) = \psi(\vec{r}).$$

Since this holds for every ψ , $\Pi^2 = \hat{I}$. □

Theorem 6.2 (Eigenvalues of the Parity Operator). [thm:parity-eigenvalues] The eigenvalues of the parity operator Π are ± 1 .

Proof. If $\Pi |\lambda\rangle = \lambda |\lambda\rangle$, then $\Pi^2 |\lambda\rangle = \lambda^2 |\lambda\rangle$. By theorem 6.1, $\Pi^2 = \hat{I}$, so $\lambda^2 = 1$, hence $\lambda = \pm 1$. □

Theorem 6.3 (Parity Operator is Hermitian). [thm:parity-hermitian] The parity operator Π is Hermitian, i.e. $\Pi^\dagger = \Pi$.

Proof. For any state $|f\rangle, |g\rangle$, we have

$$\langle f | \Pi g \rangle = \int_{-\infty}^{\infty} f^*(x) \Pi g(x) dx = \int_{-\infty}^{\infty} f^*(x) g(-x) dx = \int_{-\infty}^{\infty} (f(-x))^* g(x) dx = \langle \Pi f | g \rangle$$

Notice that $\langle f | \Pi g \rangle = \langle \Pi f | g \rangle \implies \Pi = \Pi^\dagger$. So Π is Hermitian. □

Theorem 6.4 (Parity Operator is Unitary). [thm:parity-unitary] The parity operator Π is unitary, i.e. $\Pi^\dagger = \Pi^{-1}$.

Proof. Since $\Pi^\dagger = \Pi$, and according to theorem 6.1, $\Pi \Pi = \Pi \Pi^\dagger = \hat{I}$, so $\Pi^\dagger = \Pi^{-1}$, and Π is unitary. □

Theorem 6.5 (Parity Operator is Discrete Symmetry). [thm:parity-discrete] The parity operator Π is a discrete symmetry.

Proof. The parity operators form a group:

- Set $G := \{\hat{I}, \Pi\}$.
- Operation $\cdot$: operator composition.
- Identity $e := \hat{I}$.
- Inverse $\Pi^{-1} := \Pi$ (by theorem 6.1, parity is self-inverse).

This is a discrete group with two elements, so parity is a definition 2.5. □

Theorem 6.6 (Position Operator After Applying Parity). [thm:position-under-parity] Applying Π to the position operator $\hat{x}$ (see remark 1.1) gives

$$\Pi^\dagger \hat{x} \Pi = -\hat{x}.$$

Proof. **TODO** □

Theorem 6.7 (Momentum Operator After Applying Parity). [thm:momentum-under-parity] Applying Π to the momentum operator $\hat{p}$ gives

$$\Pi^\dagger \hat{p} \Pi = -\hat{p}.$$

Proof. **TODO** □

Theorem 6.8 (General Operator After Applying Parity). [thm:operator-under-parity] For any operator $Q(\hat{x}, \hat{p})$ built from $\hat{x}$ and $\hat{p}$,

$$\Pi^\dagger Q(\hat{x}, \hat{p}) \Pi = Q(-\hat{x}, -\hat{p}).$$

Theorem 6.9 (Angular Momentum is Parity-Invariant). [thm:angular-momentum-under-parity] $\Pi^\dagger \hat{\vec{L}} \Pi = \hat{\vec{L}}$.

Proof. Apply theorem 6.8 to $\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$:

$$\Pi^\dagger (\hat{\vec{r}} \times \hat{\vec{p}}) \Pi = (-\hat{\vec{r}}) \times (-\hat{\vec{p}}) = \hat{\vec{r}} \times \hat{\vec{p}} = \hat{\vec{L}}.$$

□

Theorem 6.10 (Parity of Atomic Eigenstates). [thm:parity-atomic-eigenstates] For an definition atomic eigenstate $|\psi_{n\ell m}\rangle$,

$$\Pi |\psi_{n\ell m}\rangle = (-1)^\ell |\psi_{n\ell m}\rangle.$$

Proof. **TODO** □

Theorem 6.11 (Laporte's Rule). [thm:laporte-rule] For atomic eigenstates, the electric-dipole matrix element vanishes when $\ell' + \ell$ is even:

$$\langle n'\ell'm' | \vec{r} | n\ell m \rangle = 0 \quad \text{if } \ell' + \ell \text{ is even.}$$

Proof. **TODO** □

7 Time-Reversal Symmetry

Definition 7.1 (Time-Reversal Operator). [def:time-reversal-operator] The *time-reversal operator* $U(t)$ is defined by

$$U(t)\Psi(x, a) = \Psi^*(x, a + t)$$

Theorem 7.1 (Exponential Form of the Time-Reversal Operator). [thm:time-reversal-exponential] The *time-reversal operator* can be written as:

$$U(t) = e^{it\hat{H}/\hbar}$$

Where we assume the H (Hamiltonian) is independent of time.

Proof. **TODO** □

Definition 7.2 (Heisenberg Picture). [def:heisenberg-picture]